Homework 3
MAC Sub-Layer

Release Date 1404/2/26
Deadline 1404/3/7

Theoretical Problems

This assignment is designed to provide you with deeper insights into the concepts of the MAC Sub-Layer. If you score more than 50% of the total points, any additional points will be awarded as bonus.

Problem 1 (15 pts)

Using the Slotted ALOHA protocol for the MAC channel with the following specifications, fully illustrate and explain the packet transmission diagram. Also, calculate the final packet loss during this transmission interval. **Note:** The retransmission time for each node is assumed to be fixed. You can see these times as T_A, T_B and T_C.

Packet arrival times for the nodes:

- $T_A = 2$, Node A: $\{0, 1.8, 3.9\}$
- $T_B = 3$, Node B: $\{0.5\}$
- $T_C = 2$, Node C: $\{0.7, 4.1\}$

Problem 2 (15 pts)

In a slotted ALOHA network, 100 communication stations use a channel with a data rate of 100 Mbps. The size of each frame is 100 bits. If each station produces 1000 frames per second and sends them out:

a) calculate the throughput of the system in bits/sec.

b) If Pure ALOHA were used instead of Slotted ALOHA in this system, what would be the throughput?

c) Interpret the results and explain Why does Slotted ALOHA have higher throughput than Pure ALOHA?

Problem 3 (10 pts)

Explain whether the best choice is to select the lowest value of p in a p-persistent CSMA protocol in order to reduce the chance of collision and increase the probability of successful transmission. If yes or no, explain your answer in detail. You can use Figure 1 for your answer.

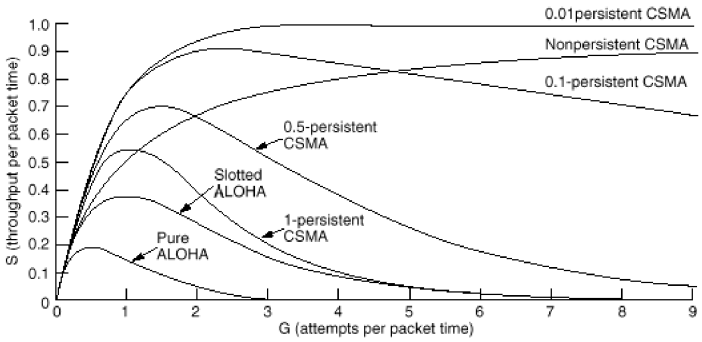


Figure 1: Throughput of different MAC protocols.

Problem 4 (15 pts)

A large population of ALOHA users generates 50 requests per second, including both original transmissions and re-transmissions. Time is slotted in units of 40 milliseconds.

(a) What is the chance of success on the first attempt?

(b) What is the probability of exactly k collisions and then a success?

(c) What is the expected number of transmission attempts needed?

Problem 5 (30 pts)

Consider the network topology shown in Figure 2, where all switches and hubs initially have empty address tables. The following sequence of frames is sent through the network:

- **Frame 1**: A $\rightarrow$ E
- **Frame 2**: E $\rightarrow$ D
- **Frame 3**: C $\rightarrow$ A
- **Frame 4**: G $\rightarrow$ H
- **Frame 5**: D $\rightarrow$ A
- **Frame 6**: B $\rightarrow$ G
- **Frame 7**: A $\rightarrow$ D
- **Frame 8**: G $\rightarrow$ C

a) Trace the Backward Learning algorithm step-by-step for each frame transmission. For every switch, show how its MAC address table updates after each frame.

b) For each frame transmission, identify the output port(s) used by the switches based on their learned tables. Justify your answers.

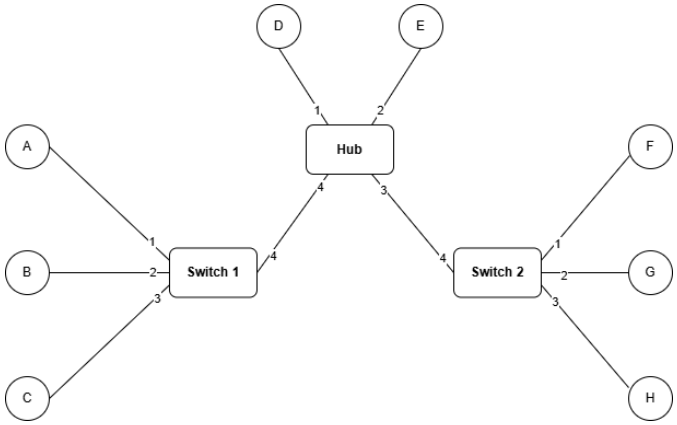


Figure 2: Sample Network Topology for Backward Learning Demonstration.

Problem 6 (15 pts)

A 1 km long, 10 Mbps CSMA/CD (Carrier Sense Multiple Access with Collision Detection) LAN (not IEEE 802.3) has a propagation speed of 200 m/μsec. Repeaters are not allowed in this system. Data frames are 256 bits long, including 32 bits of header, checksum, and other overhead. The first bit slot after a successful transmission is reserved for the receiver to capture the channel in order to send a 32-bit acknowledgment frame. What is the effective data rate, excluding overhead, assuming that there are no collisions?

Submission

Please submit your answers to the problems in a `.pdf` file named **DN_HW3_studentID.pdf**, replacing `studentID` with your actual student ID.